A Glanceable Visual Notation for Jazz Performance Routines

One round-by-round strip for drummers, rhythm section and front line

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Abstract. Jazz bands agree their routines in rehearsal: the order of choruses, who solos, where the band drops out or stops. Existing music encodings represent the order of sections but not these roles, and corpora of structural annotations record roles only after the fact. I propose one visual notation of the routine that every member of the band can read at a glance while performing, including players who may not read the lead sheet, such as drummers and vocalists. It is a strip of arrows, one per round, drawn under a tune's chord grid. Drop-outs, solo order and feel (groove, texture and dynamics) each get a fixed place and a distinct mark, following research on preattentive visual features. The notation is a proposal: it has not been tested with players. I specify a within-subjects study to test it, and outline the coding of existing routines as further research.

Keywords: jazz, performance routine, head arrangement, music notation, information visualisation, rehearsal

1 Introduction

A jazz performance has two layers of structure. The *form* belongs to the composition: a 12-bar blues, a 32-bar AABA song. The *routine* belongs to a particular band's performance: the intro, the number of choruses, who sings, who solos and for how long, where the band stops, how the tune ends. Berliner (1994, chs. 11–12) describes how such arrangements are decided in rehearsal and filled in through shared conventions.

An unwritten routine depends on every player remembering it, which is weakest with a new member, a stand-in, or a tune not played for months. A written routine helps only if it can be read between two phrases, while the eyes are also needed for the music. Sentences of remarks under a lead sheet are a poor fit, because each must be read word by word. The routine also matters most to the players least served by the lead sheet. A drummer has no melody or chords to follow, and a vocalist may sing from the lyrics without reading the chart, yet both need the routine.

The aim is therefore a *universal* notation: one strip that the drummer, the rest of the rhythm section and the front line all read, instead of a lead sheet for some and a private form chart for others. I propose a strip of arrows, one per round, designed for the songbook of the Red Jackets Jazzband, a traditional-jazz band. The paper covers the visual form only; how a routine is entered or stored is out of scope.

2 Related work

Encodings. Conventional notation marks the order of sections with rehearsal letters, repeats, endings and *D.S.*, coda and *Fine* jumps. Digital encodings carry the same information: the ABC standard lets a tune header list the play order of lettered parts with repeat counts (Walshaw, 2011); MusicXML 4.0 encodes repeats, endings and jumps, and its `time-only` type restricts an element to particular passes through a repeat (W3C Music Notation Community Group, 2021); MEI's `<expansion>` element lists the performed order of sections (Music Encoding Initiative, n.d.). All represent *order*. None has a concept of a soloist, an instrument sitting out, or a change of feel.

Annotation corpora. SALAMI (Smith et al., 2011) holds over 2,400 structure annotations of nearly 1,400 recordings, with similarity, function and instrumentation in separate layers. JAAH (Eremenko et al., 2018) annotates meter, structure and chords for 113 jazz tracks. JSD (Balke et al., 2022) segments 340 jazz recordings into over 3,000 segments, each listing its active instruments and marking the soloist and the accompaniment. The Weimar Jazz Database (Pfleiderer et al., 2017) annotates transcribed solos with chorus and form structure and per-solo metadata, including a rhythm feel. These corpora show that roles and feel can be annotated systematically, and JSD's soloist–accompaniment pairing and the per-solo rhythm feel correspond to two of the marks proposed here. But they are *descriptive*: made after the fact, time-stamped to a recording, and built for computational analysis. A routine is *prescriptive* and must be read during performance.

Practitioners' form charts. Drummers have their own, informal answer: a one-page chart of the song's whole form with only a few bars of drum notation, outlining the sections with their length in bars ("Verse – 8") plus short cues such as a groove, a hit or an energy level (e.g., Ellis, 2025; Bergamini, 2021; Ficalora, n.d.-a). Ficalora's Drum Chart Builder app lets the drummer map song form, grooves and hits (Ficalora, n.d.-b), and the trade press offers similar guidance (Lentz, 2019). Beyond drums, the Nashville number system gives the

whole band a compact, section-by-section chart for players who do not read notation (Williams, 2012). A blog post by Hein (2012) draws songs as colour-coded blocks along a bar timeline, a visual precedent for the strip's part boxes that records only section order, length and type. A web template by Scott (n.d.) is a contrast: a planning aid for the writer that says nothing about who plays in performance. Drum charts are personal and count bars; the strip is shared by the whole band and leaves bar counts to the tune's known form.

Reading at a glance. Feature-integration theory holds that simple visual features are registered early and in parallel across the visual field, while combinations of features require serial, focused attention (Treisman & Gelade, 1980). Healey & Enns (2012) survey how visualisation design uses such preattentive features to make a target found without search. For a routine, this argues for answers marked by a distinct shape in a fixed position rather than embedded in prose. Eye-movement research on music reading is much less developed than research on reading text (Madell & Hébert, 2008), and its methods vary widely (Puurtinen, 2018). I am not aware of any study of how musicians read routine information during performance.

3 Design

The target task is a player answering *"what do I do this round?"* in one glance. Six requirements follow: **R1** each kind of answer has a fixed position and a mark unlike the others (Treisman & Gelade, 1980; Healey & Enns, 2012); **R2** each fact appears in one position only; **R3** marks are ones musicians already read (plus and minus, numbered order, repeat counts, the crescendo hairpin); **R4** a round with nothing unusual adds no marks, so unusual rounds stand out; **R5** meaning is carried by position, typography and signs, not colour, so a printed copy loses nothing; **R6** nothing requires reading pitches, chord symbols or instrument-specific notation.

3.1 The strip

The strip runs left to right under the chord grid, one chevron-shaped arrow per round (Figure 1). Inside each arrow are the round number, one box per part played (A, B, Intro) and a repeat count ("4x"); an ellipsis box marks a round that lasts until a cue. Rounds playing the same part share a shade of grey. Under the arrow come the *lead* (who carries the round), underlined; one "+ instrument" line for each instrument that joins; and any other remarks on a single line. Three further questions get their own regions: **who** plays and who does not (lines under the arrow), **how** the beat goes (in, above and below the arrow), and everything else (the remarks line). "No rhythm" answers *who*, so it is a drop-out, not a feel.

Drop-outs. A "− instrument" line sits beside the "+ instrument" lines, a matched pair of signs (R3). Strikethrough was rejected because it reads as a correction. Joins and drop-outs apply to their own round only, so any arrow can be read without tracing the rounds before it. This mirrors JSD's per-segment pairing of soloist and accompaniment (Balke et al., 2022), expressed as a difference from the full band.

Solo order. A solo round shows a numbered sequence under its lead, for example $1\,\mathsf{sax/tbn} \rightarrow 2\,\mathsf{tpt} \times 2 \rightarrow 3\,\mathsf{tpt}$. One solo is one pass through the round's parts, the jazz chorus. A slash marks alternates (whichever is present), and "each ×1" fixes the length but not the players. The lengths sum to the repeat count in the arrow, so the plan can be checked at a glance.

Feel. In band usage "feel" covers three independent things, each with its own region and scope (Table 1). A groove usually holds for a whole tune and changes at most once, so it is drawn only at a change; drawing it on every arrow would violate R4. Stop time, breaks and "softly behind the vocals" end with their round. Half feel and double feel are relative to the groove in force, so they count as texture. A texture or dynamic limited to part of a round names that part ("BREAKS in the 2nd B"). Labels come from a short fixed list per axis, which can be translated on the page; a label outside the list is shown as written.

Dynamics as height. Dynamics are drawn as the height of the arrow rather than as letters such as *p* and *f*: a soft round is a lower arrow and a loud round a taller one; a crescendo widens from left to right, the shape of the hairpin players already read (R3); normal level adds nothing (R4). Arrows are centred on one line, so the number and part boxes stay in the same place whatever the height (R1). More sound maps to more ink, and length is among the more accurately judged graphical encodings (Cleveland & McGill, 1984). Only three levels are used, so each differs clearly from its neighbours without a scale.

3.2 One strip for every role

The strip is universal (R6): every answer is in one of the fixed regions, so no role needs its own chart (Table 2). No region belongs to one role; the drummer and the bass player read the same groove label, and the horns and the vocalist the same lead line. One gap remains for the drummer: the strip gives no bar counts, which drummers' own charts do.

4 Proposed evaluation

The central claim, that the strip is read faster and more accurately than written remarks, needs an experiment. I specify one here; it has not been run.

Hypotheses. **H1:** questions about drop-outs, solo order and feel are answered faster with the strip than with the same information as written remarks. **H2:** accuracy on those questions is higher. **H3:** questions about part, lead and joins are answered no slower. **H4:** the gain holds for every role, and is at least as large for players who do not read the lead sheet as for those who do.

Table 1. The three axes of feel.

Axis	Region and mark	Scope	Labels
Groove: the rhythm section's beat pattern	above the arrow: ♩ and label	until changed; drawn only at a change	two-beat, four, march, second line, shuffle, straight, latin, dirge, hymn
Texture: how much of the beat is played	foot of the round, small capitals	one round	stop time, breaks, light, vamp, pedal, half feel, double feel
Dynamics: level	arrow height: lower for soft, taller for loud; widening for build	one round	soft, loud, build

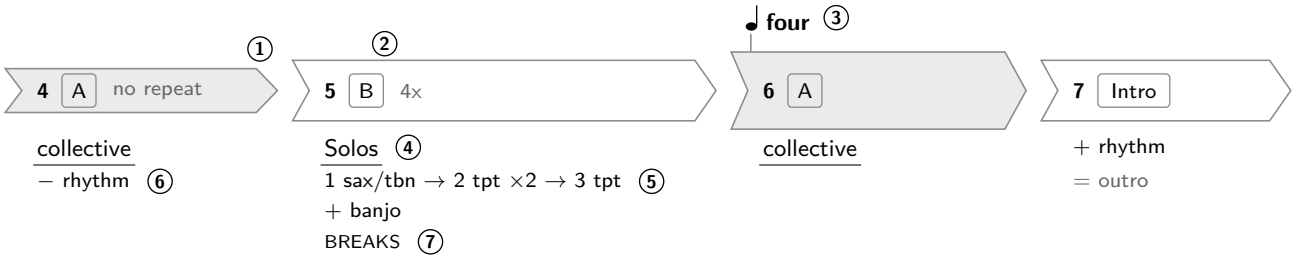


Figure 1. The strip for rounds 4–7 of an invented *Dippermouth Blues* routine. Callouts (annotations, not part of the notation): ① dynamics as arrow height, here soft (round 4) and loud (round 6); ② parts and repeat count; ③ groove, drawn only where it changes; ④ lead; ⑤ solo order with each soloist's length; ⑥ joins (+) and drop-outs (−), one round only; ⑦ texture in small capitals, one round only.

Table 2. What each role asks of the routine, and where it finds the answer.

Role	Typical questions	Where on the strip
Drummer	Where are we in the form? Does the beat change? Stop time or breaks? How loud? Who solos next?	part boxes, groove label, texture line, arrow height, solo order
Rhythm section (bass, banjo, piano)	Two-beat or four? Stop time, vamp or pedal? Do I sit out? How loud?	groove label, texture line, − lines, arrow height
Front line (horns)	Do I lead, play or rest? When is my solo? Who joins? How loud?	lead, +/− lines, solo order, arrow height
Vocalist	Which rounds are mine? Who plays behind me?	lead, + lines

Design and task. A within-subjects design with two conditions: (a) remarks as plain text below the music; (b) the strip. Participants are members and stand-ins of the band, expected $n = 8$–12, classed in advance by role and by whether they read the lead sheet while playing. Routines are counterbalanced across conditions. Each trial shows a routine briefly (about 2 s, to be set in a pilot) and asks one question drawn from Table 2, for example "In round 5, do you play?" or "Which round is loudest?" Response time and accuracy are analysed with mixed-effects models with crossed random effects for participants and routines (Baayen et al., 2008). Over a few rehearsals before and after adoption, interruptions caused by questions about the routine can also be counted. Eye tracking during performance would show where players look, but its methods vary widely and must be fixed in advance (Puurtinen, 2018). With so few participants, only large effects are detectable.

5 Discussion

Prescriptive versus descriptive. The corpora in Section 2 describe what a recording did, at the resolution of seconds. A routine states what a band intends, at the resolution of a round, so the strip shows no times; timing within a round is a short qualifier ("from the 2nd line").

Scope. The distinction between a sticky groove and per-round texture and dynamics is the part most likely to transfer to other repertoires. The label lists are less likely to: they were chosen for a traditional-jazz repertoire and not checked against other bands' usage. The strip shows the plan that cues steer, not who gives them; it shows one routine per tune, with alternatives left to remarks; and it gives no bar counts.

Further research. The notation should be applied to real routines. Coding a band's existing routines round by round against the marks would show how often each is needed, which remarks cannot be classified without the band's input, and whether the label lists suffice. I have not done this; a second coder would be needed to report agreement.

Limitations. The design is untested with players; the readability claim rests on perception research. The related-work review is selective, not systematic: "not

aware of" means only that I did not find such a study. That drummers and some vocalists do not read the lead sheet is an assumption, which H4 tests. Arrow height could be misread as importance or duration rather than level.

6 Conclusion

Routine information that standard encodings do not represent can be drawn in a compact strip: who sits out, as minus lines paired with joins; who solos and for how long, as a numbered sequence; and how the band plays, split into groove, texture and dynamics, each with its own place and scope. Because none of it needs notes or chord symbols, the same strip can serve every role. The next steps are implementing the strip, running the study above, and coding existing routines against it.

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